



IMAGE DEBLURRING

Pengolahan Citra Digital

OUTLINE

- Introduction
- Inverse filtering
 - Suffer from noise amplification
- Wiener filtering
 - Tradeoff between image recovery and noise suppression

INTRODUCTION

Where does blur come from?

- Optical blur: camera is out-of-focus
- Motion blur: camera or object is moving

Why do we need deblurring?

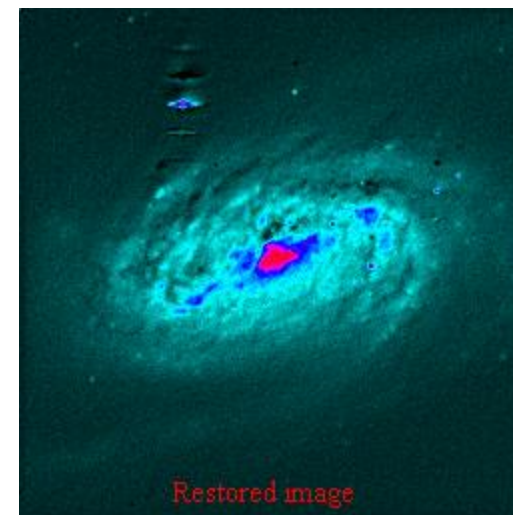
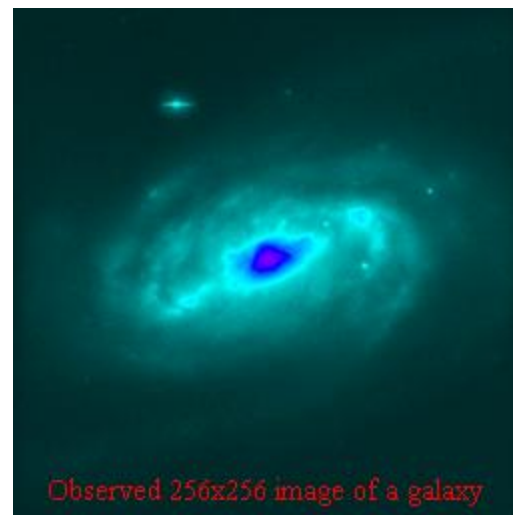
- Visually annoying
- Wrong target for compression
- Bad for analysis
- Numerous applications

APPLICATION (I): ASTRONOMICAL IMAGING

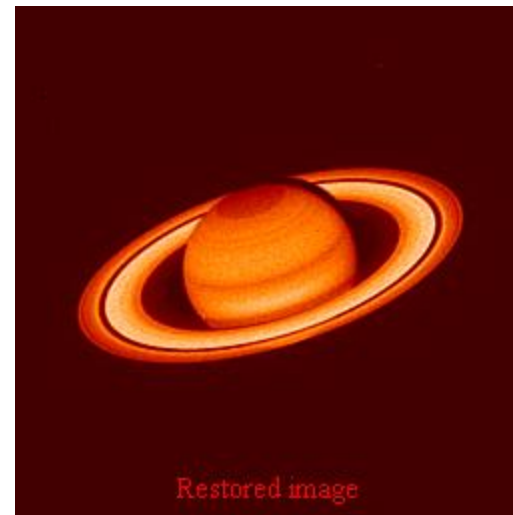
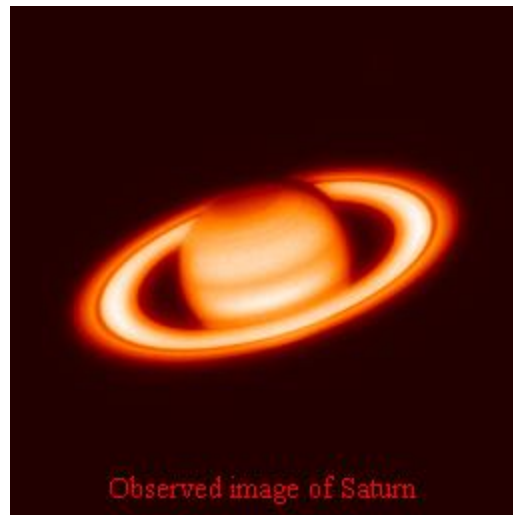
- The Story of Hubble Space Telescope (HST)
 - HST Cost at Launch (1990): \$1.5 billion
 - Main mirror imperfections due to human errors
 - Got repaired in 1993



RESTORATION OF HST IMAGES



ANOTHER EXAMPLE



THE REAL (OPTICAL) SOLUTION



Before the repair



After the repair

APPLICATION (II): LAW ⁸

ENFORCEMENT

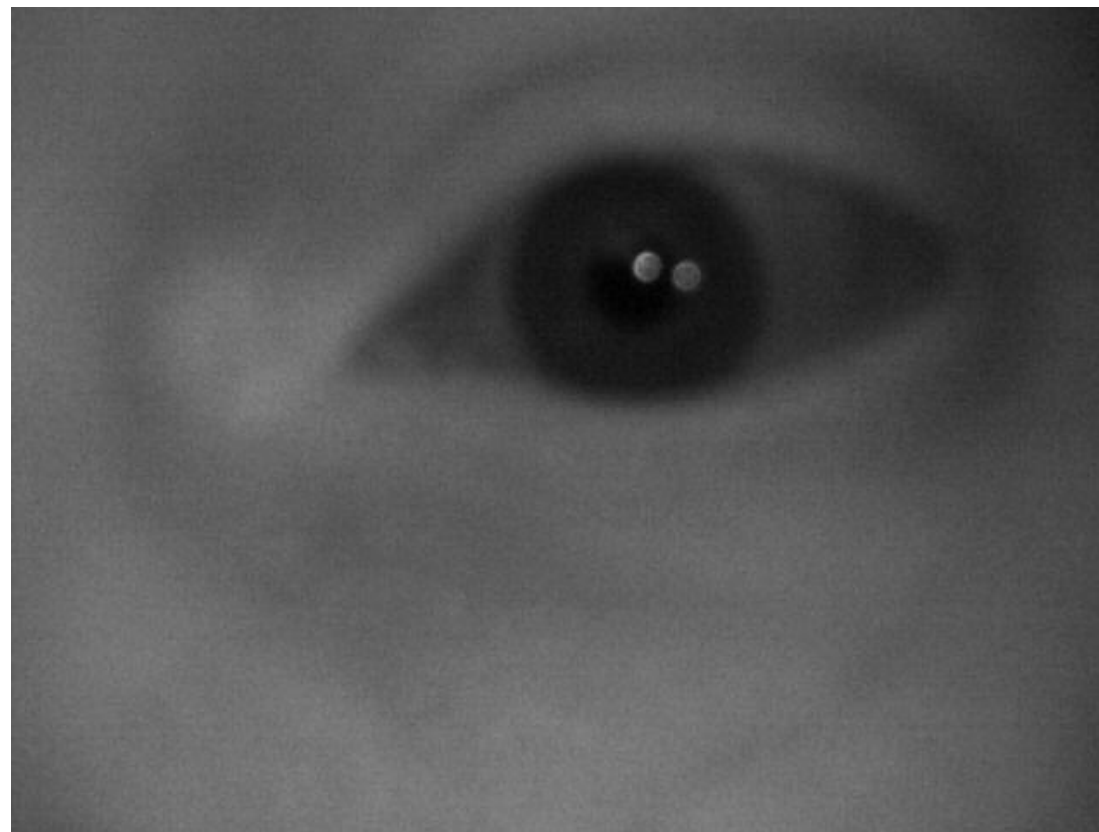


Motion-blurred license plate image

RESTORATION EXAMPLE



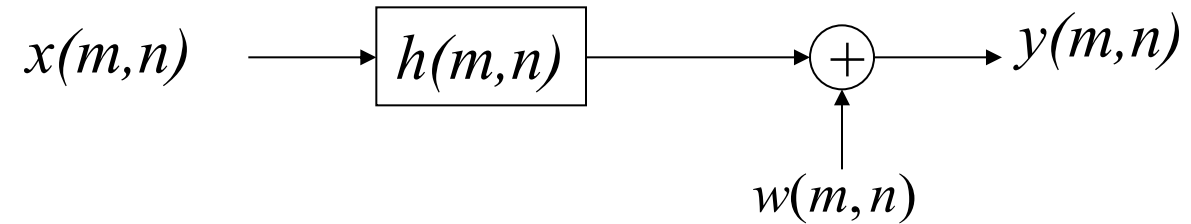
APPLICATION INTO BIOMETRICS



out-of-focus iris image

MODELING BLURRING PROCESS

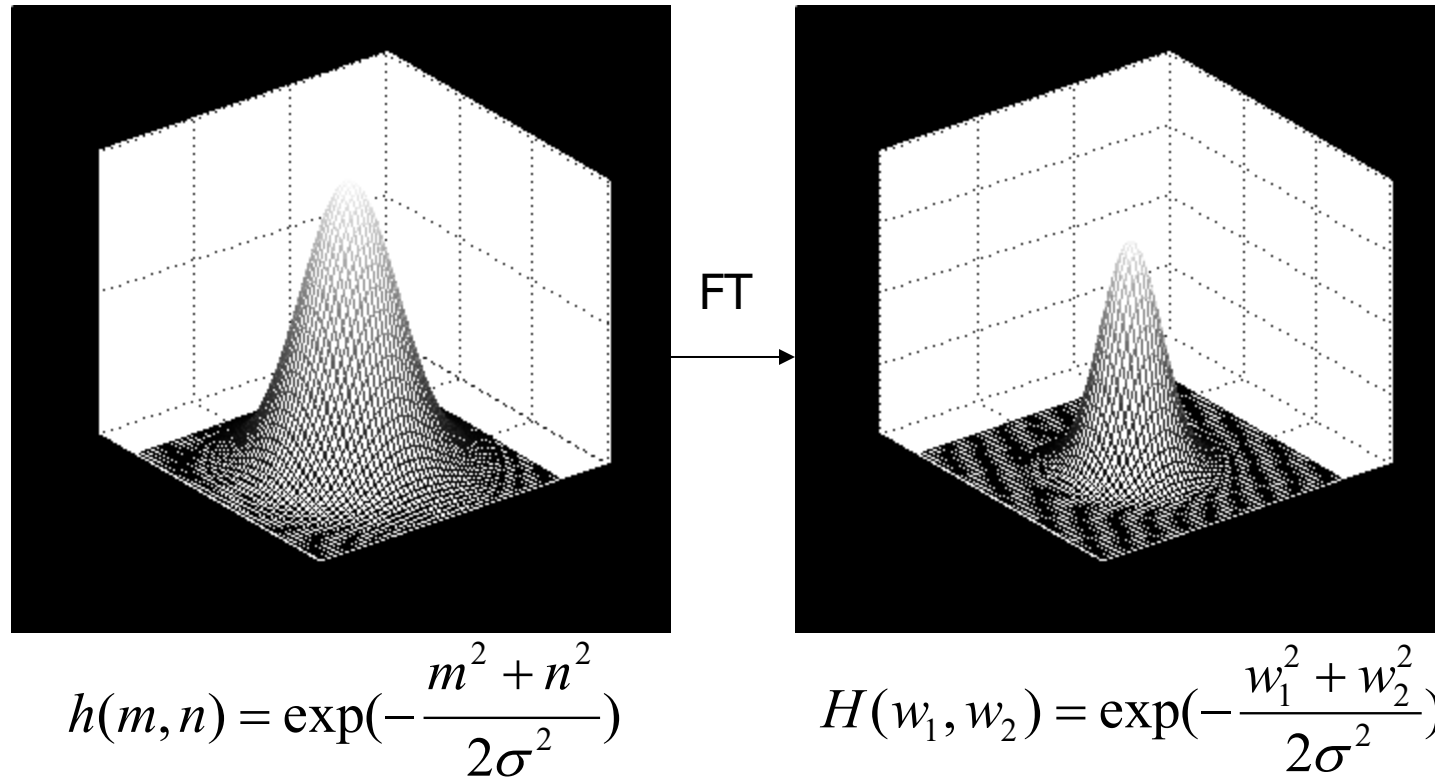
- Linear degradation model



$h(m, n)$ blurring filter

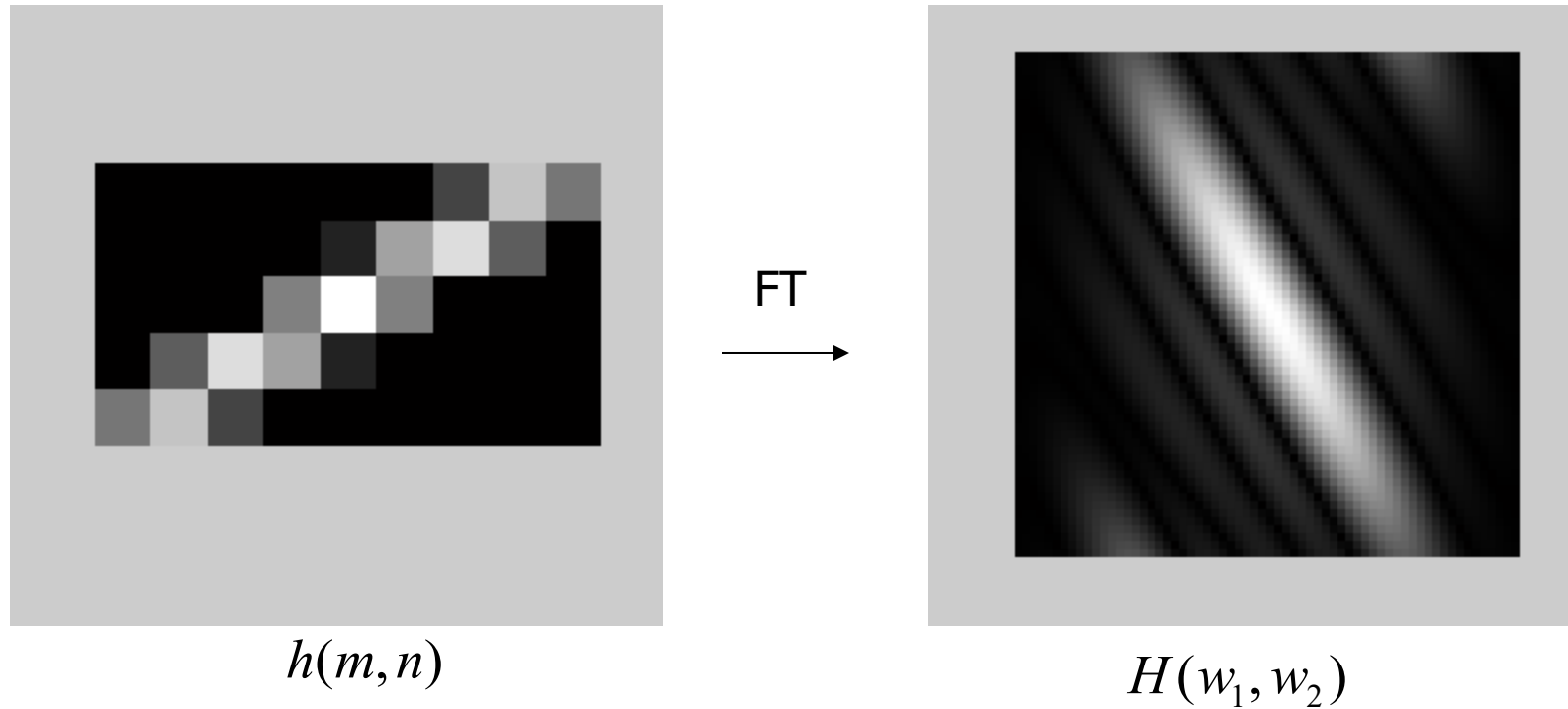
$w(m, n) \sim N(0, \sigma_w^2)$ additive white Gaussian noise

BLURRING FILTER EXAMPLE



Gaussian filter can be used to approximate out-of-focus blur

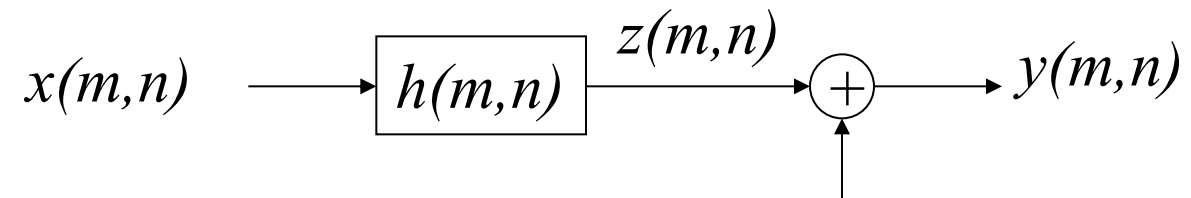
Blurring Filter Example (Con't)



MATLAB code: `h=FSPECIAL('motion',9,30);`

Motion blurring can be approximated by 1D low-pass filter along the moving direction

THE CURSE OF NOISE



$$w(m,n) \sim N(0, \sigma_w^2)$$

Blurring SNR

$$BSNR = 10 \log_{10} \frac{\sigma_z^2}{\sigma_w^2}$$

IMAGE EXAMPLE

 $x(m,n)$ 

BSNR=40dB



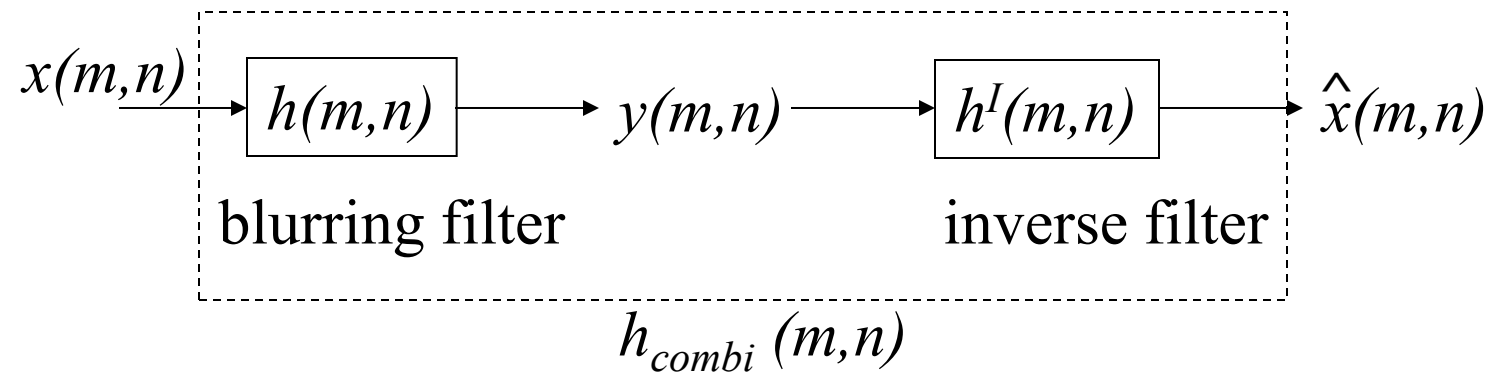
BSNR=10dB

$h(m,n)$: 1D horizontal motion blurring $[1 \ 1 \ 1 \ 1 \ 1 \ 1 \ 1]/7$

BLIND VS. NONBLIND DEBLURRING

- Blind deblurring (deconvolution): blurring kernel $h(m,n)$ is unknown
- Nonblind deconvolution:
blurring kernel $h(m,n)$ is known
- In this course, we only cover the nonblind case (the easier case)

Inverse Filter

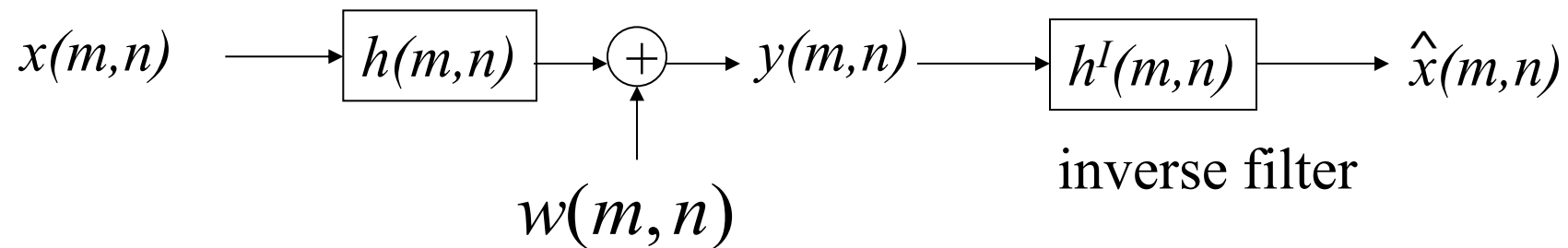


To compensate the blurring, we require

$$h_{combi}(m,n) = h(m,n) \otimes h^I(m,n) = \sum_{k=-\infty}^{\infty} \sum_{l=-\infty}^{\infty} h^I(m-k, n-l) h(k,l) = \delta(m,n), \forall(m,n)$$

$$H^I(w_1, w_2) = \frac{1}{H(w_1, w_2)}$$

INVERSE FILTERING (CON'T)



Spatial:

$$\hat{x}(m, n) = y(m, n) \otimes h^I(m, n) = (x(m, n) \otimes h(m, n) + w(m, n)) \otimes h^I(m, n)$$

Frequency:

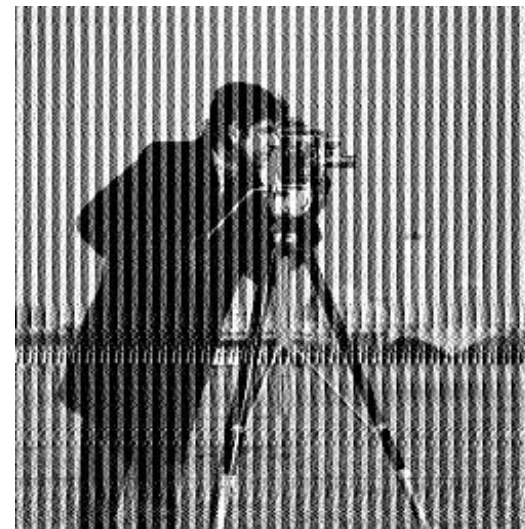
$$\hat{X}(w_1, w_2) = Y(w_1, w_2)H^I(w_1, w_2) = \frac{X(w_1, w_2)H(w_1, w_2) + W(w_1, w_2)}{H(w_1, w_2)}$$

$$= X(w_1, w_2) + \frac{W(w_1, w_2)}{H(w_1, w_2)} \longrightarrow \text{amplified noise}$$

Image Example



motion blurred image
at BSNR of 40dB



deblurred image after
inverse filtering

Q: Why does the amplified noise look so bad?

A: zeros in $H(w_1, w_2)$ correspond to poles in $H^I(w_1, w_2)$

Pseudo-inverse Filter

Basic idea:

To handle zeros in $H(w_1, w_2)$, we treat them separately when performing the inverse filtering

$$H^{-}(w_1, w_2) = \begin{cases} \frac{1}{H(w_1, w_2)} & |H(w_1, w_2)| > \delta \\ 0 & |H(w_1, w_2)| \leq \delta \end{cases}$$

Image Example



motion blurred image
at BSNR of 40dB



deblurred image after
Pseudo-inverse filtering
($\delta=0.1$)

NORBERT WIENER (1894-1964)



The renowned MIT professor Norbert Wiener was famed for his absent-mindedness. While crossing the MIT campus one day, he was stopped by a student with a mathematical problem. The perplexing question answered, Norbert followed with one of his own: "In which direction was I walking when you stopped me?" he asked, prompting an answer from the curious student. "Ah," Wiener declared, "then I've had my lunch"

Anecdote of Norbert Wiener

Wiener Filtering

Also called Minimum Mean Square Error (MMSE) or Least-Square (LS) filtering

$$H_{mmse}(w_1, w_2) = \frac{H^*(w_1, w_2)}{|H(w_1, w_2)|^2 + K}$$

constant

Example choice of K:

$$K = \frac{\sigma_w^2}{\sigma_z^2}$$

noise energy

signal energy

K=0 → inverse filtering

Image Example



motion blurred image
at BSNR of 40dB



deblurred image after
wiener filtering
($K=0.01$)

IMAGE EXAMPLE (CON'T)



$K=0.1$



$K=0.01$



$K=0.001$

Constrained Least Square Filtering

Similar to Wiener but a different way of balancing the tradeoff between

$$H_{mmse}(w_1, w_2) = \frac{H^*(w_1, w_2)}{|H(w_1, w_2)|^2 + \gamma |C(w_1, w_2)|^2}$$

Example choice of C:

$$C(m, n) = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

$\gamma=0 \rightarrow$ inverse filtering

Laplacian operator

IMAGE EXAMPLE



$\gamma=0.1$



$\gamma =0.01$



$\gamma =0.001$